

The Lexical Typology of Sensory Perception

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Keywords

sensory language, language of perception, lexical typology, semantic typology, sensory hierarchy, perception verb hierarchy

Abstract

Sensory language has fascinated researchers, as it is here that meaning most clearly straddles biology and culture. Since the seminal work on color vocabularies, typologists have attempted to describe and explain worldwide linguistic patterns in this area. One proposal that has captured the attention of many is the idea that there is a hierarchy of the senses that can account for phenomena as diverse as lexicalization patterns, frequency of use, diachronic stability, order of acquisition, and more. We argue, to the contrary, that a universal sensory hierarchy is no longer tenable. Emerging crosslinguistic data do not support a single hierarchy of the senses that is applicable across distinct linguistic and psycholinguistic properties. This does not mean we must abandon sensory language typology altogether. Alternative methods for identifying crosslinguistic regularities—such as semantic maps—show considerable promise. Moreover, while the preoccupation with a sensory hierarchy has led to an overly narrow research focus, moving beyond it opens up new avenues of research in this area. Future research has potential to uncover new patterns of sensory language structure and use across diverse languages, account for their distribution over cultures, and deepen our understanding of how language interfaces with cognition.

Implicational hierarchy: a type of generalization over languages, such that the presence of one item implies the presence of others

Sensory hierarchy: a ranking of the senses that reflects the relative importance of each sense in biology, cognition, or language

1. UNIVERSAL HIERARCHIES IN THE LANGUAGE OF PERCEPTION

The seminal book by Berlin & Kay (1969) on color vocabulary established a foundational benchmark for lexical typology, particularly for the study of the language of perception. Against the backdrop of what they termed “extreme linguistic relativity,” Berlin and Kay sought to demonstrate that there are in fact semantic universals. They collected primary data from speakers of 20 languages; for each language, they first established the basic color vocabulary using a set of linguistic and psycholinguistic criteria, and then examined the mapping of color words onto an external grid of Munsell color chips to determine the referential range of each word, as well as its best exemplar. In addition, they collated color vocabulary from written sources, bringing the total number of languages considered to just under 100. On the basis of these data, they concluded that there is a total universal inventory of 11 basic color terms within any language. A language could have as few as two color terms; critically, however, as languages added color terms to the inventory, there was a fixed hierarchical sequence to the emergence of color terms, so that an implicational hierarchy summarized worldwide patterns (**Figure 1a**). For example, if a language had a term for green, it also had terms for white, black, and red.

This study inspired a swath of like-minded endeavors to uncover semantic universals, in particular to identify sensory hierarchies. While Berlin and Kay focused on uncovering systematicities within a perceptual domain, others looked for regularities across perceptual modalities. An influential study by Viberg (1983)¹ collected data on basic perception verbs for five sense modalities

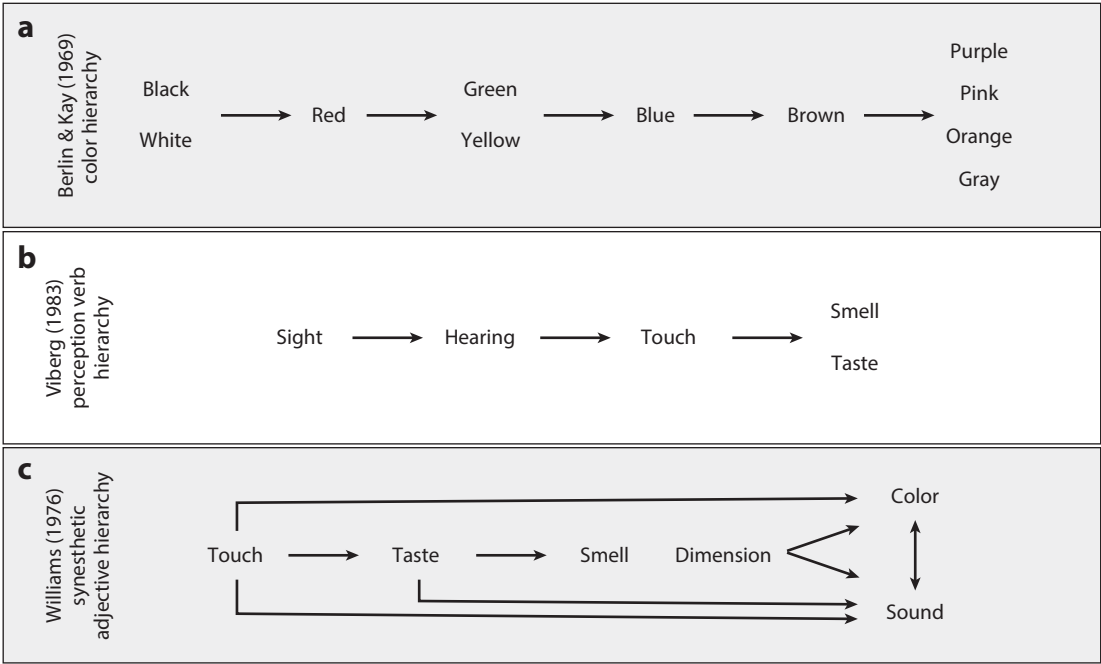


Figure 1 (a) Berlin & Kay’s (1969) color implicational hierarchy. (b) Viberg’s (1983) perception verb hierarchy. (c) Williams’s (1976) synesthetic adjective sensory hierarchy. Panel c adapted with permission from Williams (1976).

¹This paper was published also as a book chapter (Viberg 1984).

(sight, sound, touch, taste, and smell) in 53 languages, using a combination of questionnaire data and bilingual dictionaries. Viberg proposed his own sensory hierarchy to account for lexicalization patterns across languages (**Figure 1b**). He observed, for example, that “higher” senses, such as vision (left side of **Figure 1b**), were more likely to be lexicalized using simple, monomorphemic verbs, while “lower” senses, like smell (right side of **Figure 1b**), showed a greater tendency to be formally complex. The hierarchy also captured the possibility of lexical splits, with languages more likely to lexicalize a distinction between agentive and nonagentive perception for verbs denoting modalities at the top of the hierarchy (e.g., ‘look’ versus ‘see’) and less likely to do so for lower modalities. Viberg also found that, across languages, semantic extensions proceeded from higher to lower along the hierarchy, as observed in word formation, where perception verbs combine with other elements to denote the sense modalities beneath them (e.g., hear + odor > ‘smell’). In later studies, Viberg (2001, 2008) noted other differences in, for example, text frequency of perception verbs (“higher” senses have higher frequency), lexical differentiation (vision was most likely to be encoded in a dedicated verb), and language learning (vision verbs were acquired earlier in first and second language acquisition).

Separately, Williams (1976) attempted to characterize diachronic regularities in meaning change for sensory experience adjectives, specifically with respect to meaning transfer within the sensory domain. Based primarily on dictionary data from 100 English adjectives, supplemented with additional data from Indo-European and Japanese, Williams’s hierarchy stipulated that touch meanings extend to other senses, including vision, but not the other way around (**Figure 1c**). He concluded that sensory meanings transfer “from the physiologically least differentiating, most evolutionary primitive sensory modalities to the most differentiating, most advanced, but not vice versa” (Williams 1976, pp. 464–65), an idea advanced earlier by Ullmann (1948).²

Various scholars subsequently noted that there is a near-converse patterning over Viberg’s and Williams’s hierarchies, where perception verb meanings extend down the hierarchy whereas adjectives extend upward (Evans & Wilkins 2000, Strik Lievers 2015). The same hierarchical approach to sensory language has been applied to other word classes, like ideophones (Akita 2009, Dingemanse 2012).

Aside from the shared presumption that patterns of meaning across languages can be characterized by a hierarchy, another parallel across studies is the idea that human biology is the main driver of semantic patterns. The color hierarchy has been attributed to “pan-human perceptual universals” (Berlin & Kay 1969, p. 109); in contrast, perception hierarchies are claimed to reflect the relative dominance of the five senses in human perception, which are “strongly grounded in human biology and general cognition” (Viberg 2008, p. 127), according to evolutionary, ontogenetic, and neurophysiological considerations (see also Williams 1976). More recently, researchers have argued that language reflects perceptual and neuroanatomical constraints, so the dominance of vision in language may be explained by the heavier weighting of visual information in multi-sensory integration (e.g., Winter et al. 2018) and volume of cortical activation across senses, with vision surpassing the rest (e.g., Reilly et al. 2020).

In Section 2, we take a closer look at sensory hierarchies in light of the accumulating linguistic and psycholinguistic evidence, by examining whether there are universals in perception language frequency (Section 2.1), perception verb lexicalization (Section 2.2), codability (Section 2.3), and acquisition (Section 2.4). In Section 3, we consider the categories over which sensory hierarchies have been built, showing that neither psychophysiology (Section 3.1) nor crosslinguistic data

Codability: the ease with which a concept can be expressed in language, according to lexical availability, grammatical simplicity, or processing efficiency

²Based on an analysis of nineteenth-century poetry, Ullmann (1948, p. 387) stated that transfers of sensory meaning “tend to mount from the lower to the higher reaches of the sensorium, from the least differentiated sensations to the most differentiated ones, and not vice-versa.”

Sensory property: a specific quality or attribute (e.g., color, shape, size) perceived through a sensory modality

Sensory modality: main channel through which the world is perceived (e.g., sight, hearing, touch, taste, smell)

Type: a unique word (or unit), regardless of how many times it occurs

Token: each individual occurrence of a word (or unit)

Efficient codability: a psycholinguistic construct referring to the ability to name an attribute quickly and consistently (within and across speakers)

(Section 3.2) support a five-sense model. We argue that it is important to distinguish between sensory properties and modalities (Section 3.3), before outlining possible avenues of research that follow from these considerations (Section 3.4). In Section 4, we discuss new frameworks for understanding the language of perception, going beyond unidimensional hierarchies to multidimensional approaches (Section 4.1). We also consider explanations for typological patterns in the language of perception beyond biology, including linguistic, communicative, and cultural considerations (Section 4.2), and discuss the implications of variable linguistic coding of perception on cognition (Section 4.3). We conclude in Section 5.

2. QUESTIONING SENSORY HIERARCHIES

The typological literature has conceived of sensory hierarchies in two distinct ways. On the one hand, they are presented as a descriptive summary of a typological distribution. Viberg (1983, p. 137), for example, in describing the semantic extensions of perception verbs, writes that “a verb having a basic meaning belonging to a sense modality higher (to the left) in the hierarchy can get an extended meaning that covers some (or all) of the sense modalities lower in the hierarchy.” On the other hand, the hierarchy is presented as a functional explanation for that same distribution, as suggested by statements such as “the hierarchy restricts the patterns of polysemy that are possible” (Viberg 2001, p. 1297; see also Evans & Wilkins 2000). Of course, to account for sensory hierarchies it is necessary to ground their explanation in factors external to the phenomenon itself; otherwise, the reasoning is circular (Dąbrowska 2016, Gibbs 2007, Winter 2019). However, searching for external explanations is necessary only insofar as the original empirical generalizations are robust.

Establishing whether there is a universal sensory hierarchy requires at least two things. First, the hierarchy should be observed across a wide range of unrelated languages. Second, it should be clear what range of phenomena the hierarchy covers. Sensory hierarchies could be reflected in various ways. One possibility suggested in the literature is that there are asymmetries in lexical coding—more differentiated vocabulary (types and tokens) for vision, followed by sound, touch, taste, and smell. The asymmetry may also be reflected in lexicalization patterns, for example, in how word forms extend their meanings across senses. It has also been suggested that there should be differences in psycholinguistic measures, such as ease of naming sensory stimuli (i.e., efficient codability), and that the hierarchy may be reflected in language acquisition. This is by no means an exhaustive list, but consideration of these factors will illustrate that sensory hierarchies must be abandoned as a way to think about the language of perception.

2.1. Perception Language Frequency

It is claimed that token frequencies of basic perception verbs reflect the sensory hierarchy: Verbs of vision are more frequent than those of hearing, which in turn are more frequent than those of touch, and so on (e.g., Viberg 2001, 2008). This hypothesis was supported by an initial study of Indo-European languages (Viberg 2001). A later study found that not only verbs but also adjectives and nouns referring to vision in English had higher frequency than words referring to other sensory modalities (Winter et al. 2018). This has been the case across registers and language varieties since the 1800s. In fact, type frequencies (more distinct terms) pattern with token frequencies (how often each type is used), such that there are more unique words to refer to vision in English and fewest for smell and taste (Winter et al. 2018).

A different study of perception verbs in conversational speech across 13 diverse languages (from nine language families and six continents) found general support for visual dominance but more variation among the other senses (San Roque et al. 2015). Vision verb forms were

statistically more frequent than others for 10 of 13 languages, and more frequent in 11 of 13 languages when comparing verb meanings.³ Visual dominance in perception verb token frequencies is further supported by corpus studies of individual languages (Floyd et al. 2018, Holmer 2021, Krishna et al. 2022, Tchamtourian & Vamling 2005, Viberg 1993) and by psycholinguistic studies of type frequencies—although these data come primarily from major languages, raising questions about generalizability (Chen et al. 2019; Filipović Đurđević & Popović Stijačić 2019; Lynott & Connell 2009, 2013; Lynott et al. 2020; Miceli et al. 2021; Miklashevsky 2018; Morucci et al. 2019; Repetto et al. 2023; Speed & Brybaert 2022; Speed & Majid 2017; Vergallito et al. 2020; but see Zhong & Ahrens 2023).

San Roque et al. (2015) found that verbs of hearing show a strong tendency to come second in frequency of talk (in 12 of 13 languages). The relative ranking of touch, taste, and smell showed more crosslinguistic variability across the board, with no clear evidence of touch outranking smell and taste. In fact, Semai (Aslian) showed a higher frequency for smell than hearing, in keeping with the observed importance of olfaction in Aslian languages and cultures (Burenhult & Majid 2011, Kruspe & Majid 2023, Majid & Burenhult 2014, Majid & Kruspe 2018, Tuck-Po 2021, Tufvesson 2011, Wnuk & Majid 2014). In Cha'palaa, smell verbs were more frequent than touch verbs, a finding that has been replicated using a corpus almost 100 times larger than that of the original study (Floyd et al. 2018).

Moreover, Cha'palaa has 15 basic smell terms, which appear more frequently in everyday conversation than do smell terms in English or Quechua (Floyd et al. 2018). This finding is in line with claims that the sensory lexicon is communicatively efficient, making more distinctions (tokens) in places where there is greater need (usage frequency) (Winter et al. 2018). It is surprising, then, that within sensory modality there are more distinctions among types of negative smells but the frequency of positive smell terms in discourse is higher (Siahaan 2022), a finding that calls for further study. Taken together—and contrary to claims of a universal sensory hierarchy—the evidence demonstrates crosslinguistic differences in type and token frequencies of perceptual language.

2.2. Perception Verb Lexicalization

It is claimed that lexicalization processes are constrained by the perception verb hierarchy, so senses at the top of the hierarchy, like 'see' and 'hear', are more likely to have a dedicated verb than those lower down (Viberg 2001, 2015). In keeping with this generalization, Avatime (Kwa, Niger-Congo) has a dedicated verb *mɔ* for 'see' and a single perception verb *nu* 'hear/feel/taste/smell' for all nonvisual senses (van Putten 2020). Inversions of this pattern are supposedly rare (e.g., a dedicated verb for 'taste' at the same time as a verb covering 'see', 'hear', and 'feel'). This claim regarding colexification received empirical support from an initial analysis of the Database of Cross-Linguistic Colexifications (CLICS²), which found that nonvisual senses tend to colexify with one another (Georgakopoulos et al. 2022).

However, CLICS² featured a geographically unbalanced language sample, and not all perception verbs were recorded, casting some uncertainty on the typological robustness of this finding. In addition, previous studies had not examined the lexicalization of nonvisual senses to test whether they pattern hierarchically as predicted. To address this question, Norcliffe & Majid (2024a) collated the full paradigm of perception verb lexicons in a balanced sample of 100 languages. They found that vision showed a strong typological bias to be lexically differentiated from other perception meanings, contrary to previous claims that "verbs of visual perception easily spread into other

Perception verb hierarchy: a ranking of perception verbs (e.g., *see, hear, feel*) related to their linguistic behavior, including frequency and lexicalization patterns

Colexification: expression of two or more concepts in a single word

³For Cha'palaa (Barbacoan) and Chintang (Sino-Tibetan), although not statistically significant, the difference was also in favor of vision.

Dislexification: the inverse of colexification; distinct concepts are expressed in distinct words

Lexical codability: the availability of a simple word (or lexeme) within a language to express a specific attribute, for example, a particular color or odor

sensory meanings” (Huumo 2010, p. 57; see also, e.g., Jędrzejowski & Staniewski 2021, Traugott & Dasher 2002). Beyond vision, lexicalization patterns were not consistent with the predicted sensory hierarchy either: Languages were, in fact, less likely to encode hearing and touch meanings with a dedicated verb compared with smell and taste. In a striking pattern, unrelated sign languages exhibit a preference for dislexification in the domain of perception, with distinct signs for each perceptual modality, typically articulated close to the relevant sensory organ (Zeshan & Palfreyman 2019). In summary, the results do not support the claim of a fixed hierarchy of senses constraining lexicalization patterns.

2.3. The Codability of Perception

Rather than test the posited sensory hierarchy by examining lexical codability (see Section 2.1), we can measure efficient codability instead (Levinson & Majid 2014). If visual information is the most accessible to language, followed by hearing, touch, taste, and smell, then this should be evident in ease of naming, too. One study (Majid et al. 2018) compared speakers from 20 diverse languages—including 3 unrelated sign languages—for how consistently speakers described sensory stimuli of various visual properties, sounds, tactile textures, tastes, and smells. Only English matched the predicted sensory hierarchy better than chance: Colors and shapes showed the highest naming agreement, followed by sound, then taste, touch, and finally smell. The other 19 languages had 13 different sensory hierarchies. For example, Kata Kolok (a village sign language of Bali) had the highest codability for taste, followed by vision, smell, and finally touch;⁴ in contrast, Dogul Dom (Niger-Congo) had the highest codability for touch, taste, color, sound, smell, and then shape (Majid et al. 2018, Majid & Levinson 2026).

Other studies have challenged the hierarchy by pitting smell against vision in sensory naming tasks, again finding cultural relativity (e.g., Majid & Burenhult 2014, Majid & Kruspe 2018). English speakers demonstrate higher naming agreement for vision than smell, as predicted by the hierarchy, but speakers of both Jahai and Semaq Beri (Aslian, Austroasiatic) show equal naming agreement for smell and vision. Intriguingly, Semelai (also an Aslian language) shows the same asymmetry in naming as English, suggesting that it is not the language family or geography that is driving this effect. Instead, it seems that culture underpins naming patterns—the two hunter-gatherer communities show no asymmetry between smell and sight, whereas the non-hunter-gatherer communities favor vision over olfaction. Yet again, we see no evidence of a universal hierarchy of the senses.

2.4. The Typological Prevalence Hypothesis in Acquisition

Some scholars have proposed that the sensory hierarchy is reflected in the acquisition of perception verbs (Bloom et al. 1989; Viberg 1993, 2001): Children will learn sight verbs, then hearing verbs, before touch, taste, or smell verbs because of a natural human propensity to orient to visual and auditory meanings, which is reinforced through patterns of language use. To test this idea, San Roque et al. (2024) investigated the learning trajectories of basic perception verbs in children learning English, using data from parent questionnaires and corpora of children’s spontaneous speech. They found that children acquire visual perception verbs (‘see’ and ‘look’) earliest, at 12–23 months, but touch rather than audition verbs come second, in both age of acquisition and production frequency. Later, at 24–35 months, frequency rates shift, and audition begins to align with reported adult frequencies. In the earliest age window, then, the learning trajectory does

⁴Auditory stimuli were not tested for deaf participants.

not follow the sense hierarchy. Drawing on data from another 11 languages in Wordbank (Frank et al. 2017), San Roque et al. (2024) replicated the early acquisition of touch verbs, even in a few non-Indo-European languages.

The touch-before-hearing finding could be interpreted within the framework of embodied theories of word meaning (Barsalou 1999, Pulvermüller 1999), which hold that language use is related to specific bodily experiences. From an embodiment perspective, and in keeping with the traditional Piagetian view of the developmental course of sensorimotor development, the prominence of touch over hearing verbs could reflect the early relevance of the tactile modality for infants, for whom much of the experienced world is explored through the hands. However, it also reflects infants' linguistic experience: Child-directed speech shows a higher frequency of touch verbs compared with hearing verbs (in contrast to adult–adult speech) in the earliest age bracket. Critically, this use of touch vocabulary is driven by talk about behavior regulation in the speech of caregivers toward young infants, especially prohibitions against touching, rather than a focus on tactile sensation per se (San Roque et al. 2024). These findings raise fascinating questions about how individual sensory experience interacts with linguistic input to scaffold language learning (Bedny et al. 2012, Campbell et al. 2025, Hauptman et al. 2025, Landau & Gleitman 1985). More generally, these observations underscore the need to look beyond the body—to language use and sociocultural practices—to explain the semantic packaging of perception.

Etic: An etic approach uses universal, standardized categories and measures based on the viewpoint of an external observer

3. GOING BEYOND FIVE SENSES

Sensory hierarchies rest on the assumption that there are five separate senses, each linked to a different sense organ (eyes, ears, nose, etc.), that can be arranged in a set of linear dominance relations. The five senses could be kept as “a useful fiction” for the purposes of linguistic theorizing, a “first-pass generalization that may form an adequate starting point for investigating language and perception” (Winter 2019, p. 13). But we are at a stage now where this fiction is not so useful. The implicational hierarchies considered above presume an etic grid with universal semantic categories that can be extracted from biology. However, the notion that there are five senses that are fully distinct departs from both scientific understanding and crosslinguistic data.

3.1. Psychophysiology Does Not Support a Five-Sense Model

In addition to the external senses, there are physiological systems for sensing internal sensations such as pain and temperature (interoception) and the location and movement of one's body (proprioception). The eminent psychophysicist Marks (1978, p. 2) wrote: “It is hardly disputable that there are seven different external senses—more or less,” the “more or less” conditional on uncertainty about how to treat the sense of touch (see also Longo 2024). There are also deep cross-modal interactions between the senses at behavioral and neural levels. Consider smell and taste. Sensory scientists distinguish the ability to detect airborne volatile molecules by sniffing (orthonasal olfaction) versus through the mouth (retronasal olfaction) (Majid 2024). In fact, smell—specifically, retronasal olfaction—is responsible for most of what the layperson describes as taste, giving rise to the complex experiences of fruity, smoky, and meaty flavors (Spence 2015).

The overall number of perceptual senses could be enumerated in different ways: If senses are identified with sense organs, then perhaps there are five. If, on the other hand, senses are identified according to the type of stimulus, we may count three instead (light = vision, mechanical = hearing and touch, chemical = smell and taste), and if they are identified by receptor, there are hundreds (there are four types of photoreceptors in the retina for vision, but around 400 unique olfactory receptors). Of course, receptor types are not available to conscious awareness and, therefore, would not be expected to be encoded linguistically. Nevertheless, this example illustrates the general

point that biology is more complex than presumed, and insofar as sense hierarchies are argued to be directly grounded in human biology, this is problematic.

Emic: An emic approach applies categories and meanings that reflect the perspective of the specific language and culture being studied

3.2. The Crosslinguistic Data Do Not Support a Five-Sense Model Either

Typically in semantic typology, when an etic grid is to be composed, the theorist first aims to capture the exhaustive, nonredundant set of semantic distinctions made crosslinguistically and uses it to map individual language patterns (Evans 2011). The posited five senses for the perception domain are not sufficient on these grounds either. Some Australian languages lexically distinguish internal bodily sensation from external touch, for example. Arrernte's set of basic perception verbs includes both *anpeme* 'touch, feel by touch, feel (rough, smooth, etc.)' and *welheme* 'have a (propri-oceptive) feeling, feel (cold, sick, hot, and so on), feel something doing something to you' (Evans & Wilkins 2000). In South America, Tariana (Arawakan) makes a lexical distinction between *-rena* 'feel a general internal state, be in a good or bad state' and *-hima* (causative *-himeta*) 'feel cold, sick, hot, someone touching you' (Aikhenvald & Storch 2013). Meanwhile, in Africa, in Anlo-Ewe (Atlantic Congo) the verb *seselelame* is glossed as 'feeling in the body' and appears to cover all perceptual senses, while specific terms distinguish *agbagbaɔɔɔ* 'vestibular sense', *azɔlizɔɔɔ* 'kinesthetic sense', *nulele* 'tactile sense', and more (Geurts 2002). These examples suggest that a five-sense etic grid is not sufficient for capturing crosslinguistic distinctions; finer-grained distinctions are needed.

Equally, the emic perspective problematizes the status quo. Going back to Anlo-Ewe, Geurts (2002, p. 49) argues that speakers "did not seem to experience or conceptualize perceptual processes as restricted to five discrete channels," giving examples such as "hearing in the skin" or "hearing odor." Along similar lines, Hausa is said to "recognize only two senses compared with our five," *gani* 'to see' and *ji* 'to perceive nonvisually' (Ritchie 1991, p. 195). Similarly, Forth (2023) argues that a binary distinction in Ngadha-Lio languages corresponds to a major ontological distinction between, on the one hand, humans and animals who can be seen and, on the other hand, spirits who are perceived nonvisually via sound, smell, or touch. Such examples hint at alternative conceptualizations of the senses. At the same time, it is important to recognize the role of both conventionalized constructions and in-context inference in distinguishing the senses (Evans & Wilkins 2000, van Putten 2020). Beneath these apparent binary distinctions, there could be distinct senses at play. Careful linguistic and cultural analysis is required to adjudicate these possibilities (see also Wierzbicka 1996).

3.3. Sensory Modalities and Sensory Properties

Some sensory hierarchies focus on sensory modalities, but others include sensory properties. So, while Viberg's (1983) perception verb hierarchy relies strictly on a five-sense model, Williams's (1976) adjective hierarchy includes color, not the visual modality, and he also introduced the notion of dimension (Figure 1). Later implicational hierarchies operate over still other properties, such as movement and pattern (e.g., Dingemanse 2012). While there are distinct phenomenal qualities associated with external stimuli as they activate different sensory organs (consider the cooling effect of menthol on the skin versus its minty aroma), different sensory modalities can inform us about the same physical features—for example, both geometric shape and surface texture can be perceived by sight and touch, unlike color or temperature, whose perception is restricted to a single channel. And some dimensions of experience apply to (nearly) all perceptual modalities—consider intensity and duration.

These examples highlight additional complexities in attempting a typology of sensory language: To what extent is any hierarchy generalizable over different types of properties perceived

across different sensory organs? Consider the study of efficient codability by Majid et al. (2018), described in Section 2.4. To test the posited sensory hierarchy, participants described sensory stimuli such as colors, shapes, and textures. Colors and shapes were apprehended visually and textures haptically. Notably, the two visual domains did not always pattern in the same way—in Tzeltal, for example, color was highly codable but shape came at the other end of the language hierarchy. Whether the same sensory hierarchy would emerge if shapes were perceived by hand and textures by eye is not known.

3.4. Opening the Door to New Questions

If we step away from the five-sense model, many questions open up about the semantic packaging of the sensorium. We can ask what elements of meaning get grouped together or distinguished, just as with any other semantic domain.

3.4.1. The spatial structuring of sensory language. Seminal research by Talmy (1985, 2000) identified two common strategies that languages use to convey motion events—either lexicalize path of motion in the main verb (e.g., Spanish *entrar* ‘move-in’) or put path in a particle associated with the verb (a “satellite”) and squeeze manner into the main verb (e.g., English *roll down/up/across*). This distinction may leak into the perception domain, too. English makes a rich set of distinctions for manner of vision (e.g., *glimpse*, *spot*, *scrutinize*) (Levin 1993), but other languages, like Maniq (Austroasiatic), lexicalize path with vision (e.g., *balay* ‘look up’, *yɔp* ‘look down’, *ciyek* ‘look sideways’) (Wnuk 2022; see also Evans & Wilkins 2000, Klein 1981), contrary to earlier claims that lexicalized visual paths do not occur (Matsumoto 2001, Slobin 2009). The extent to which other languages follow this split remains to be determined (Capelle 2020, Cifuentes-Férez 2014, Kawachi 2020). Another open question concerns the possibility of constructing complex spatial path descriptions in combination with vision verbs, as satellite-framed English and Russian seem to, but verb-framed Spanish and Turkish do not (Slobin 2009).⁵

Unrelated sign languages also encode perception spatially. For example, signs for ‘to see’ universally express movement away from the body, whereas signs for ‘to hear’ show more variability in path expression (e.g., Takashima 2019, Zeshan & Palfreyman 2019), raising questions about modality-specific spatial mappings. Similar spatial structuring has been observed in spoken languages, too. In Kadiwéu (Guaicuruan), sight and smell are lexicalized with distinct verbs that encode proximal and distal distinctions—for example, ‘see at a distance’ versus ‘see up close’ and ‘smell at a distance’ versus ‘smell up close’ (Griffiths 2002). There are also more implicit associations: A Dutch psycholinguistic study found that nouns with strong smell associations were processed faster in proximal than distal space, as if simulated as close to the body (Speed & Majid 2017). These observations suggest that spatial parameters may play an extensive role in the lexicalization of perception and in how it is processed across modalities and languages.

3.4.2. Sensory language and valence. Languages lexicalize nonsensory meaning components together with sensory ones. For example, it is not uncommon for languages to incorporate an evaluative component in perception verbs—most obviously for pain (e.g., *hurts*, *stings*, *throbs*, *aches*) (Reznikova et al. 2012), temperature (Koptjevskaja-Tamm 2015), other interoceptive states (Zhong & Ahrens 2023), and smell and taste vocabulary (Allan & Burridge 2006, Buck 1949, Classen 1993, Jurafsky 2014, Krifka 2010, Viberg 2001, Winter 2016). Oromo (Cushitic), for example, has separate verb roots for ‘taste good’ and ‘taste bad’ and ‘smell good’ and ‘smell bad’, but

⁵Curiously, the perceptual modality by which an event is experienced appears to influence language production: Motion events that are only heard elicit more path and fewer manner descriptions than those that are seen (Mamus et al. 2023).

Semantic map:

a visual representation that captures typological regularities by placing concepts or meanings in space, such that closely related meanings are close together in space

Directed graph:

a visual representation where nodes (e.g., meanings) are connected by edges that are directed (i.e., one-way)

no neutral terms (Viberg 2001). The default interpretation of *smell* is to smell bad, but *taste* is taste good, a fact Krifka (2010) puts down to more commonly experiencing bad smells but good tastes (for related explanations, see Classen 1993, Winter 2016). Perhaps relatedly, where languages lexicalize distinct types of odor, there are more terms for negative smells than positive ones (Boisson 1997, Majid 2021b). However, an experiential asymmetry is unlikely to be the root cause of these patterns. When psychologists have tracked how many odors a person perceives each day, frequently experienced odors are primarily pleasant (Mahmut et al. 2021). Unrelated languages have been reported to be in line with Krifka's smell–taste valence asymmetry (François 2021, Vanhove & Ahmed 2021); however, absence of counterexamples is not evidence of absence. A systematic study is needed.

3.4.3. Sensory language beyond the lexicon. Beyond the lexicon, sensory perception can be expressed in other areas of grammar. Some languages have dedicated means for grammatically expressing information source, known as evidentiality markers (Aikhenvald 2004), which often make distinctions on the basis of whether the information source was acquired through direct (visual) perception versus another sense modality. In some languages, perceptual meanings are encoded in demonstratives (e.g., Dyirbal *ngala-* ‘not visible; either audible or remembered’) (Evans & Wilkins 2000). Yet other languages have specialized morphemes, for example, the Tundra Nenets odorative suffix *xal’a-yə-* ‘smell of fish’ (*-yə-* ‘smell of’) (Nikolaeva 2014).

Nominal classification systems frequently reflect visual and tactile categories, such as shape and consistency, but classifiers based on auditory categories are unattested; olfactory and gustatory classifiers, although attested, appear to be rare. Cha’palaa is one language with such classifiers, where sensory quality terms are constructed by combining a closed set of abstract roots that cannot appear independently, with the smell classifier *-dyu* or taste classifier *-mbu* (Floyd et al. 2018). How the encoding of sensory information is distributed and combined across different subsystems within grammars, and how this encoding interacts with distinctions within lexicons, awaits comprehensive study.

4. NEW FRAMEWORKS FOR UNDERSTANDING THE LANGUAGE OF PERCEPTION

Unidimensional hierarchies are not sufficient for capturing the typological patterning of sensory lexicons. Even the classic color hierarchy has been substantially revised to recognize more complex crosslinguistic patterning of color lexicons (Kay et al. 2009). General principles of semantic change—for example, semantic narrowing and broadening—apply to the color domain, with color terms disappearing from lexicons and not only accruing, as suggested by the classic model (Bowern & Haynie 2017, Haynie & Bowern 2016; but see Nash 2017). The fact that general semantic change principles apply to the color domain demonstrates that factors beyond “pan-human universals” shape color vocabulary (Conway et al. 2020, Gibson et al. 2017, Wnuk et al. 2022, Zaslavsky et al. 2019). The same considerations apply to the rest of the sensory lexicon: Our frameworks must evolve to more accurately reflect and explain the typological facts.

4.1. Beyond Unidimensional Hierarchies to Multidimensional Structures

The simplest semantic map organizes meanings (nodes) in space with edges (lines). In the twentieth century, researchers attempted to synthesize a large quantity of language data by hand and unsurprisingly favored unidimensional semantic maps; however, some putative simple “hierarchies” ended up more like complex directed graphs (Viberg 1983, Williams 1976). As typologists moved to systematizing both data collection and analysis, other types of models appeared (e.g., Croft 2022, Croft & Poole 2008, Georgakopoulos & Polis 2018, Haspelmath 2003, Majid 2008,

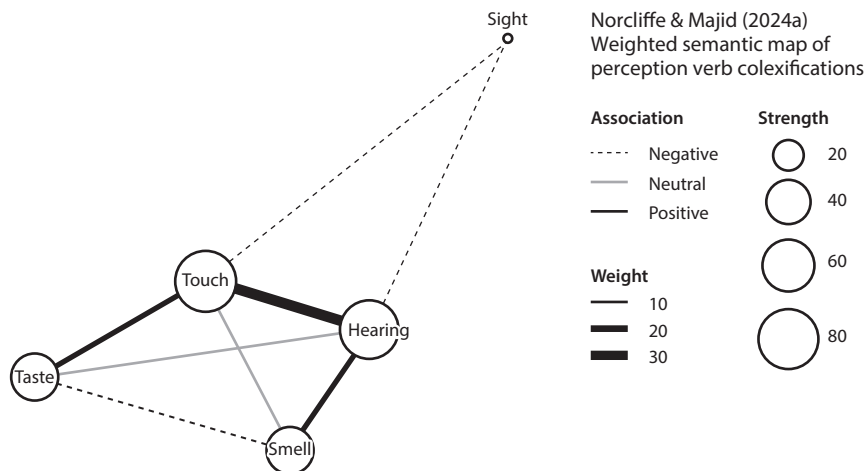


Figure 2

An example of a weighted semantic map. Each node represents a sensory modality, with the size of the circle representing its strength score (i.e., how frequently that meaning colexifies with others). Edges indicate colexifications—the thicker the line is, the more often that colexification appears crosslinguistically. Solid black lines indicate a positive association, dashed lines a negative association, and gray lines a neutral association (neither more nor less likely than chance). Figure adapted from Norcliffe & Majid (2024a).

Majid et al. 2008, Regier et al. 2013). These models reveal different ways to model semantic patterning across languages.

Using weighted semantic maps to model colexification patterns of sensory meanings in basic perception verbs, for example, Norcliffe & Majid (2024a,b) found that the visual modality is negatively associated with all other modalities (i.e., vision verbs dislexify within the perception domain). Moreover, smell and taste were also negatively associated (**Figure 2**). This finding is surprising because, as mentioned above, many researchers had concluded that vision verbs are wont to extend to other sensory meanings (e.g., Huumo 2010, Jędrzejowski & Staniewski 2021, Traugott & Dasher 2002), and the close conceptual relatedness between smell and taste would predict that these modalities share lexical forms (e.g., Viberg 1983, 2001; Winter 2016). Moreover, it appears that direct extension from ‘see’ to ‘hear’ is not possible and is attested only through semantic shifts outside and then back into the domain—for example, ‘see’ → ‘understand’ → ‘hear’ or ‘look’ → ‘pay attention to’ → ‘listen’ (Georgakopoulos et al. 2022, Norcliffe & Majid 2024a, San Roque et al. 2018). The use of semantic maps allows researchers to quantitatively test and establish such relationships.

Winter & Strik-Lievers (2025) endorse a similar conclusion in their analysis of linguistic synesthesia across languages. They found that a simple model testing for a unidimensional hierarchy was supported but that a semantic network better characterized the data. Semantic maps have also been used to represent the relationships between sensory categories depicted by ideophones (McLean 2021, van Hoey 2023). For example, a weighted semantic map applied to ideophones elicited with nonlinguistic stimuli from speakers of Japonic languages found that sound, movement, shape, and texture tend to be sources for semantic extensions (McLean 2021). Like sound and movement, shape and texture may be motivated by the ease with which they can be iconically mapped to speech (McLean 2021). Directed graphs have also been used to capture patterns of cross-modal extensions of adjectives in particular languages (Alvarado et al. 2024, Shinohara & Nakayama 2011, Strik Lievers 2015).

Weighted semantic map: a type of semantic map where nodes (e.g., meanings) are connected by edges (i.e., lines) that differ in thickness (i.e., “weight”) to convey the strength or frequency of the relationship between nodes

4.2. Beyond Biology: Linguistic, Communicative, and Cultural Considerations

Although no single sensory hierarchy is attested across all language phenomena, there are asymmetries in the lexical encoding of sensory language, including how sensory meanings are encoded in word classes and in patterns of colexification across sensory domains. These asymmetries likely arise from multiple factors, including general principles of cognition and communication, as well as language-internal factors and broader cultural influences. Disentangling these factors and their interactions is an active and ongoing area of research, as demonstrated below through a few illustrative cases.

4.2.1. Sensory meanings across word classes. Different sense modalities are preferentially encoded in particular lexical categories (e.g., nouns versus adjectives) because of a natural alignment between the properties of sensory experiences and the prototypical functions of specific lexical categories. For example, English sound concepts are most prone to being encoded as verbs, perhaps because they are inherently dynamic and involve change and movement, which align with the prototypical properties of verbs (Strik Lievers & Winter 2018). These asymmetries may, in turn, explain why certain sense modalities are more likely to occur as sources of cross-modal extensions. For example, a study of English and Italian found that adjectives are more numerous for touch than hearing, while the opposite is the case for nouns (Strik Lievers 2015). This finding could explain why sensory adjectives typically extend from touch to sound rather than from sound to touch (Ronga et al. 2012, Strik Lievers & Winter 2018, Ullmann 1948, Winter 2019). To date, studies of the differential lexical affordances of the senses have been restricted to only a handful of languages, so whether these observations are crosslinguistically generalizable is unknown.

4.2.2. Explaining patterns of colexification. In contrast, the factors influencing the colexification of sensory meanings has received considerable attention. Some have suggested that the same word or form is applied to conceptually related meanings because of a cognitive pressure for simplicity: Meanings that require less effort to associate—because they are conceptually closer—are more likely to share a word. Within the language of perception, there is already a high degree of conceptual resemblance, but nevertheless there are asymmetries. For example, the combinations touch–taste and hearing–smell are frequent in verbs, whereas the inverse pairings touch–smell and hearing–taste are rare, suggesting a conceptual association between the contact senses on the one hand and the noncontact senses on the other (Norcliffe & Majid 2024a, Viberg 1983). This association in itself is not trivial, in light of claims that the conceptualization of the senses is highly variable cross-culturally (Aikhenvald & Storch 2013, Classen 1993, Wierzbicka 1996). Notably, however, the crosslinguistically most frequent pairing in verb meaning is hearing–touch (Norcliffe & Majid 2024a), crosscutting the notion of contact. A post hoc conceptual resemblance explanation could be offered for this association: Sound can be felt as vibrations conducted through the body, and in the other direction, tactile properties such as roughness can be perceived auditorily. However, these sorts of post hoc explanations quickly become unsatisfactory. To be effective, conceptual similarity explanations must be stipulated in advance and it must be possible to falsify them. Moreover, it is unclear what the causal directionality is here—does meaning similarity give rise to shared labels, or do shared word forms drive conceptual similarity (see Section 4.3)?

Theories of communicative efficiency take a different starting point, positing that languages evolve to trade off competing pressures of informativeness (supporting communicative effectiveness) and simplicity (minimizing cognitive load). A pressure for informativeness favors one-to-one mappings between meanings and words; a pressure for simplicity favors colexification of related meanings (i.e., less lexical differentiation within a domain). This trade-off is mediated by

communicative need. Semantic categories tend to be narrower (more lexically differentiated) in high-frequency portions of a given semantic domain and broader (colexified) in low-frequency (low-need) portions of the domain (e.g., Gibson et al. 2017, Kemp et al. 2018, Zaslavsky et al. 2019). Communicative need may explain the typological tendency in the perception verb domain for visual meanings to dislexify with other sensory meanings (Norcliffe & Majid 2024a). This unimodal bias of vision verbs is not easily explainable on conceptual grounds: Vision, like hearing and smell, is a noncontact sense, so if hearing and smell tend to colexify on this basis, why not vision? From the perspective of communicative efficiency, however, vision verbs are the most frequent of all perception verbs across languages (Section 2.1), suggesting a greater need to communicate about vision across cultures, which in turn may drive languages to lexically differentiate vision from the other senses.

The deeper question is: What drives the cross-cultural tendency to talk most about visual experiences? Here we circle back to perceptual accounts, as this tendency could, as Viberg (1983) originally proposed, be “grounded in human biology”—that is, we often talk about visual experiences because, for many people, vision plays a primary role in perception (see Wang et al. 2026). Consider, though, that there may be other reasons why speakers recruit verbs of vision in speech. Visual perception verbs are frequently conventionalized for pragmatic functions, for example, to redirect a sequence of talk (Kendrick 2019, San Roque et al. 2015, Sidnell 2007), possibly because vision is the sense typically relied on for joint attention (Moore & Dunham 1995). So, the social functions of the visual modality could underlie the high frequency of visual perception verbs in speech. While this possibility does not rule out an ultimate explanation at the biological level (i.e., we draw on the visual modality for pragmatic purposes because of its perceptual dominance), it points to the need for accounts that differentiate levels of causal explanation. It also opens up the compelling possibility that communicative needs vary systematically across cultures. For example, olfactory lexicons may arise in some communities but not others because of different subsistence patterns and related cultural practices (Majid 2021a,b; Majid & Kruspe 2018), color vocabularies may be related to colored objects in the environment (e.g., Berlin & Kay 1969, Gibson et al. 2017, Rivers 1901, Wnuk et al. 2022), and shape vocabularies may be efficiently codable in communities with formal education (Majid et al. 2018).

Communicative need: how often speakers need to refer to a certain concept or idea

4.3. The Cognitive Consequences of Variable Linguistic Coding of the Senses

Variable linguistic coding of the senses raises the question of whether there may be cognitive consequences of these distinctions. This topic has been amply covered in the domain of color (Davidoff et al. 1999, Franklin et al. 2008, Kay & Kempton 1984, Kopp & Lane 1968, Maier & Abdel Rahman 2018, Winawer et al. 2007) and continues to be debated (for recent reviews, see, e.g., Lindsey & Brown 2021, Lupyan et al. 2020, Witzel 2019).

Similar questions can be asked about each of the senses and about the relative hierarchy of the senses, too. For example, languages differ in how they describe auditory pitch; languages like English and Dutch use a high–low metaphor, whereas other languages like Farsi and Turkish use a thin–thick metaphor (Eitan & Timmers 2010, Shayan et al. 2011). This linguistic variation seems to influence nonlinguistic behavior, such as singing, gesturing, and cross-modal matching (Dolscheid et al. 2013, 2020; Fernandez-Prieto et al. 2017; Holler et al. 2022), and is visible early in development (Dolscheid et al. 2023, Shayan et al. 2014), although the extent of these differences is still in contention (Parkinson et al. 2012, Starr & Srinivasan 2018). Training studies in the lab have shown that labels can influence tactile (Miller et al. 2018) and odor (Vanek et al. 2021) discrimination, but whether these effects can be found in real-world contexts and independent of other cultural variables remains to be determined.

5. CONCLUSION

Early accounts of the language of perception attempted to fit many different linguistic phenomena under the umbrella of a postulated sensory hierarchy grounded in biology. The hierarchy was to explain formal complexity, polysemy patterns, direction of derivational and semantic extensions, order of acquisition, relative frequencies, relative diachronic stability, and more. But it is questionable whether a single principle or framework could ever be expected to capture so many diverse linguistic phenomena, a point that is underscored if the goal is to explain patterns in other pockets of the sensory lexicon, too. Some explanations may indeed transcend multiple linguistic phenomena, but confirming this hypothesis requires a systematic and detailed collection of individual language data from a diverse sample of languages to enable the methodological rigor required of linguistic typology. There are many exciting avenues to pursue.

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